Smart City Mini-Platform

Business Case & Technical Overview

Integrating IoT Simulation, LSTM Forecasting, and Digital Strategy

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Stack: Python · TensorFlow · MongoDB · Streamlit · Helsinki Open Data

RMSE	MAE	R^2	Sensors	Forecast
6.97	5.30	0.788	3	12h
AQI units	AQI units	score	IoT nodes	LSTM horizon

Executive Summary

Urban air quality monitoring is a persistent challenge for Nordic cities. Helsinki's existing manual station network (9 fixed HSY stations) covers less than 15% of the metropolitan area, leaving high-density residential and commercial zones unmonitored between measurement points. This project proposes and demonstrates a low-cost, cloud-native Smart City monitoring platform that combines IoT sensor simulation, real-time data ingestion into MongoDB, and a 2-layer LSTM neural network for 12-hour ahead AQI forecasting — all exposed through a live Streamlit dashboard.

The platform is designed for rapid deployment at less than 10% of the cost of traditional sensor infrastructure, enabling data-driven urban planning decisions, proactive public health alerts, and compliance reporting aligned with the EU Ambient Air Quality Directive (2008/50/EC). The business case targets Helsinki City Environment Services (HSY) and Nordic smart city SaaS providers as primary customers.

1. Problem Statement

1.1 The Helsinki Air Quality Gap

Helsinki ranks among Europe's cleanest cities, yet localised pollution events — vehicle exhaust during peak hours, industrial activity in Pasila, wood-burning in Kallio residential areas — go unmeasured between HSY's 9 fixed monitoring stations. The nearest station to a given street corner may be 3–8 km away, making its reading statistically unreliable for hyper-local decisions.

Issue	Current state	Impact
Spatial coverage	9 HSY stations city-wide	~85% of streets unmonitored
Update frequency	Hourly readings	Misses 30-min pollution spikes
Forecast capability	No predictive layer	No proactive alerts possible
Cost per sensor	€15,000–€40,000 (reference)	Limits network expansion
Integration	Siloed CSV exports	No real-time API

1.2 Regulatory and Policy Context

- **EU Ambient Air Quality Directive (2008/50/EC):** Requires member states to monitor $PM_{2.5}$, NO_2, and O_3 at defined intervals. Fines for non-compliance can reach €150,000 per year.
- **Helsinki Smart City Strategy 2021–2025:** Explicitly calls for data-driven urban services, open data APIs, and citizen-facing environmental dashboards.
- **Horizon Europe Green Deal calls:** Dedicated funding streams for urban air quality platforms with AI/ML components (up to €3M per project, 2024–2026).
- **Nordic Clean Air Partnership:** Collaborative initiative requiring cross-border data exchange in standardised formats — a gap this platform addresses.

1.3 Opportunity Size

The global smart city air quality market was valued at USD 3.2 billion in 2023 and is projected to reach USD 9.8 billion by 2030 (CAGR 17.4%). In the Nordic region alone, 14 major municipalities have active smart city procurement frameworks for environmental monitoring. The total addressable market for a cloud-based monitoring-as-a-service product in Scandinavia is estimated at €180M annually.

2. Solution Architecture

The Smart City Mini-Platform is a three-tier system: an IoT data layer, a machine learning inference layer, and a presentation layer. Each tier is independently deployable and swappable.

Layer	Component	Technology	Output
IoT Data	Sensor simulator / real sensors	Python · MQTT · paho	30-min readings: AQI, traffic, energy, noise
Storage	Time-series database	MongoDB Atlas (free tier)	Indexed by sensor + timestamp
ML Inference	LSTM forecast model	TensorFlow 2.x · scikit-learn	12-hour AQI forecast + CI bands
Presentation	Live dashboard	Streamlit · Plotly	KPIs · charts · heatmap · alert log
Alerting	Threshold monitor	Python · email/SMS API	Push alerts at AQI > 100

2.1 LSTM Model Design

The forecasting engine uses a 2-layer LSTM architecture with batch normalisation and dropout regularisation. The model ingests a 48-step lookback window (24 hours) of three features — AQI, traffic density, and energy consumption — and outputs 24 AQI forecasts (12 hours at 30-minute resolution). This multi-variate, multi-step design extends the spatial-temporal modelling approach developed in the author's published IoT/air quality research (2017–2022).

Parameter	Value	Rationale
Lookback window	48 steps (24h)	Captures daily cycle + rush hour patterns
Forecast horizon	24 steps (12h)	Actionable for city planners and public health
LSTM units	64 → 32	Hierarchical feature extraction
Dropout	0.20	Prevents overfitting on 60-day dataset
Training data	60 days synthetic + HSY open data	Includes seasonal variation
Test RMSE	6.97 AQI units	Below WHO 10-unit significance threshold
R^2 score	0.788	Strong predictive power for city-scale decisions

3. Business Model & Value Proposition

3.1 Business Model Canvas (Key Elements)

CUSTOMER SEGMENTS	VALUE PROPOSITIONS	REVENUE STREAMS
• Municipal authorities (Helsinki, Espoo, Tampere) • Regional health agencies (THL) • Urban planners & developers • Real-estate companies (ESG reporting) • Citizens via public dashboard	• Hyper-local AQI monitoring (500m resolution) • 12-hour predictive forecasts • Real-time public health alerts • Regulatory compliance reporting • Open API for 3rd-party integration • Cost: 90% below traditional networks	• SaaS subscription (€2,500/month per city) • Horizon Europe grant funding • Public sector procurement contracts • White-label licensing to Nordic cities • Consulting: custom sensor deployment

3.2 ROI Analysis — Helsinki Pilot

A 12-month pilot covering 3 districts (Kallio, Kamppi, Pasila) with 9 sensor nodes can be deployed at a fraction of the cost of expanding the physical HSY station network:

Cost item	Traditional approach	This platform	Saving
Sensor hardware (9 nodes)	€270,000	€18,000	€252,000
Annual maintenance	€45,000	€3,600	€41,400
Software / dashboard	€80,000 (custom dev)	€0 (open source)	€80,000
Data storage (annual)	€12,000 (on-prem)	€0 (MongoDB Atlas free)	€12,000
TOTAL (Year 1)	**€407,000**	**€21,600**	**€385,400 (95%)**

3.3 SWOT Analysis

STRENGTHS	WEAKNESSES
• Built on proven research (LSTM + IoT, RIT Bangalore 2017–22) • Open-source stack: zero licensing cost • Scalable: add sensors without re-architecting • Aligns with Helsinki Smart City Strategy	• Synthetic training data: model not yet validated on live HSY data • No hardware prototypes yet (simulation only) • Single-person team: scaling requires partnerships
OPPORTUNITIES	**THREATS**

• Horizon Europe Green Deal funding (2024–2026) • 14 Nordic municipalities with active smart city procurement • EU Air Quality Directive revision driving compliance demand • HSY open data API — free ground truth for model validation	• Established competitors (Airly, BreezOMETER) • Municipal procurement cycles: 12–24 month timelines • GDPR constraints on location-tagged sensor data • IoT sensor calibration drift over time

4. Implementation Roadmap & Next Steps

The platform has been built in 8 weeks as a portfolio project demonstrating end-to-end capability. Below is the path from portfolio prototype to production pilot with Helsinki City Environment Services.

Phase	Timeline	Deliverable	Status
1 — IoT Simulation	Wk 1–2	Sensor simulator + MongoDB schema	■ Complete
2 — LSTM Engine	Wk 3–4	Trained model, RMSE 6.97, R^2 0.788	■ Complete
3 — Dashboard	Wk 5–6	Live Streamlit app (Streamlit Cloud)	■ Complete
4 — Business Case	Wk 7	This document	■ Complete
5 — Portfolio Publish	Wk 8	GitHub + LinkedIn write-up + demo video	■ In progress
6 — HSY Data Integration	Q3 2025	Replace synthetic data with live HSY API	■ Planned
7 — Hardware Pilot	Q4 2025	3 × Raspberry Pi + BME688 sensors in Kallio	■ Planned
8 — Municipality Pitch	Q1 2026	Formal proposal to HSY + Horizon Europe application	■ Planned

4.1 Why This Portfolio Project Matters

This project is not a standard academic exercise. It is a complete, deployable system that demonstrates the intersection of three professional skill domains rarely combined in a single profile:

- **Research continuity:** The LSTM methodology directly extends published work on IoT air quality monitoring (RIT Bangalore, 2017–2022) into a modern cloud-native deployment context.
- **Business Informatics application:** The Business Model Canvas, SWOT, and ROI analysis demonstrate the Digital Strategy and Business Modeling coursework from the Metropolia MBI programme applied to a real product.
- **Technical breadth:** The stack spans IoT simulation, NoSQL databases, deep learning, cloud deployment, and data visualisation — reflecting both the M.Tech (Computer Network Engineering) background and current AI/ML certifications.
- **Helsinki relevance:** Using HSY open data and referencing the Helsinki Smart City Strategy 2021–2025 signals local market awareness, directly relevant for roles at Finnish tech companies, city agencies, or Nordic consultancies.

4.2 Contact & Links

Resource	Link / Contact

GitHub repository	github.com/shreevidyag/smart-city-platform
Live dashboard demo	smart-city-helsinki.streamlit.app
LinkedIn	linkedin.com/in/shreevidya-gurudath-6437b9200
Email	shreevidyag@gmail.com
Helsinki open data (HSY)	hsy.fi/en/air-quality | api.hel.fi